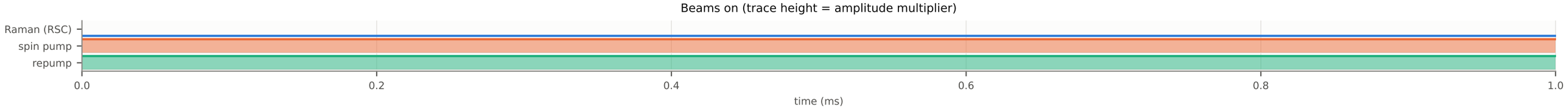
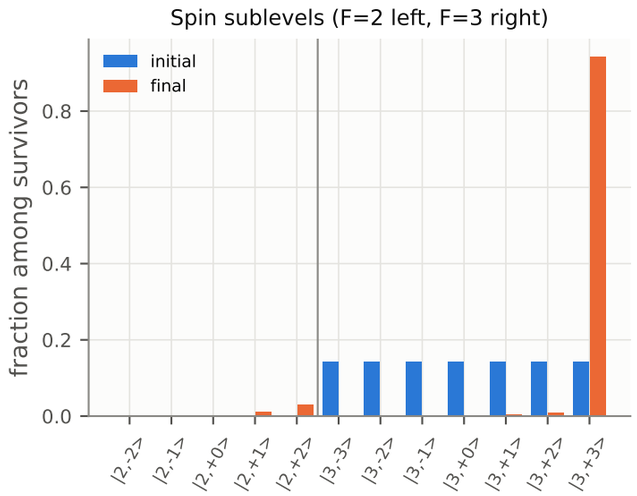
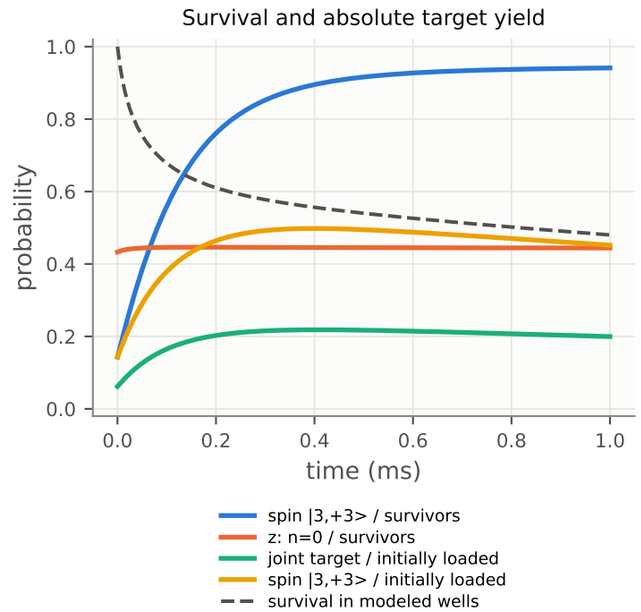
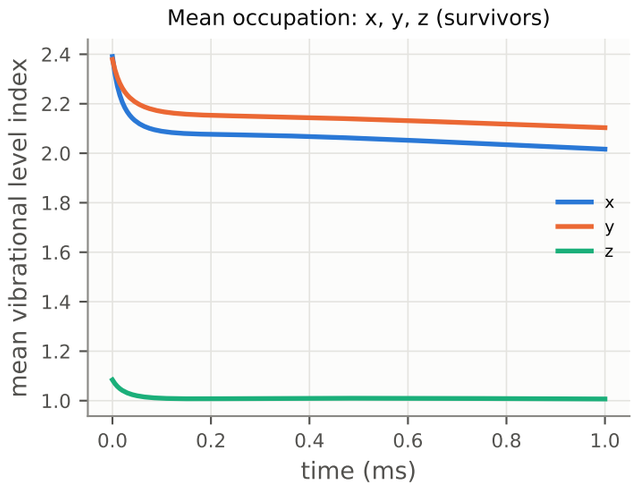
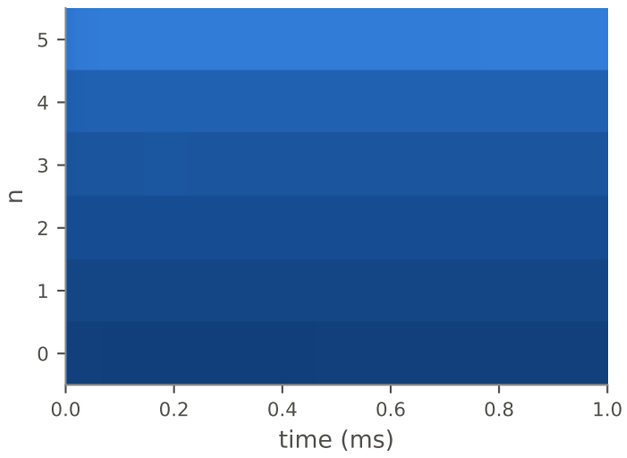
onlyOP - diagonalized atomic states - protocol custom
Model validity: INVALID — outside model validity limits

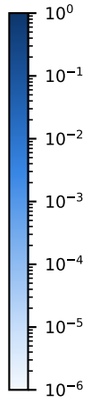


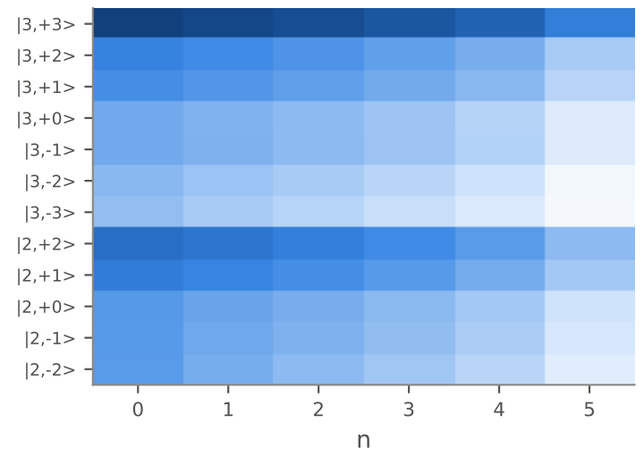


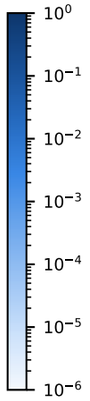


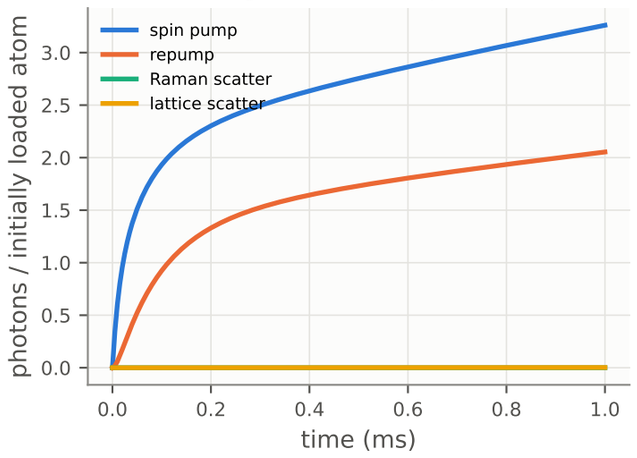


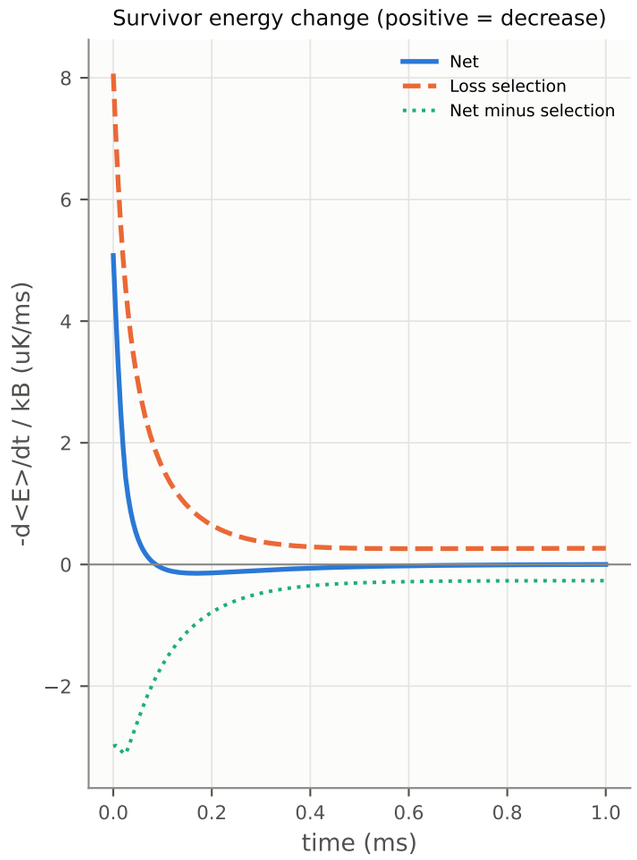













```
MODEL VALIDITY: INVALID
Target motional axis: local z; Raman light off throughout schedule.
Raman pair: A=|3,+3> <-> B=|2,+2>; P_up and target refer to A, P_down to B.
P_up=0.9411  P_n0=0.4442  P_target=0.4156  (P_up*P_n0=0.4180)
Well escape assuming no recapture: 0.5198
Axis temperature estimates, initial -> final (uK): x: 10 -> 6.13, y: 10 -> 6.71, z: 5 -> 4.52
Thermal x/y/z plots separate net energy removal from preferential-loss selection; a falling survivor temperature alone does not
establish Raman cooling.
Axis temperature proxies (uK): x=6.13, y=6.71, z=4.52
populations are per survivor: remaining in wells 0.4802 (well escape, no recapture 0.5198); initially loaded fractions: P_up=0.4519
P_n0=0.2133  P_target=0.1995
nbar (trapped atoms) 1.084 -> 1.007   T_equiv=4.52 uK [Energy-matched mixture of local bound canonical distributions, weighted by
surviving spatial populations; proxy only. None means no finite positive-temperature match.]
photons/atom: pump 3.26, repump 2.05, Raman 0
target |3,+3> scattering rate 1.05e+03 /s; quanta removed per photon None
t(P_up>=0.9) = 0.418 ms;  t(P_target>=0.5) = not reached
conservation err 4e-15; min eig 0; boundary pop unspecified; overflow 0.52
solver coherent_3d_split_ensemble, wall 14.5 s

[OK     ] magnetic field: Ground, D1, and D2 hyperfine + Zeeman eigenstates and mixed optical dipoles; F,m labels follow the B=0
branches adiabatically. Fine-structure mixing and diamagnetism omitted.
[OK     ] ground coherences: adjacent-mF Zeeman angular splitting / maximum photon rate across nodes = 45.6 (needs >>1)
[OK     ] conservation: max |survival + well loss - 1| = 4e-15
[OK     ] positivity: minimum density eigenvalue / population = 0
[OK     ] weak excitation: maximum eliminated excited fraction across nodes and spin states = 0.005052
[INVALID] motional secular: max optical rate / smallest local motional angular spacing across all axes = 2.84
[OK     ] carrier calibration: Raman light is off throughout the schedule; no carrier calibration is applied.
[WARNING] three-dimensional recoil: Joint spin/nx/ny/nz populations retained, but emission uses resolved spherical dipole channels
with factorized Cartesian marginals. Interference between emitted spherical channels and correlations of one photon's Cartesian
recoil components are neglected.
[WARNING] recoil coherence: Population-resolving scattering erases motional coherence and treats elastic spin amplitudes as
distinguishable; this is an additional approximation beyond secular averaging.
[WARNING] lattice: Separable isolated wells with Gaussian local depths; no tunnelling, coherent Raman continuum excitation, envelope
transport or recapture. Scalar trap basis with diagonal differential shifts; trap shape changes from spin-dependent shifts
neglected.
[WARNING] Stark shift approximation: Raman/trap Stark operators use diagonal matrix elements; static off-diagonal motional dressing
and trap-basis changes are omitted. Excited-state lattice shifts of the pumping transitions remain the configured common optical
shift.
[WARNING] spatial ensemble: 64 frozen Gaussian position / uniform registration samples. Refine samples and time_step_us for
convergence; samples are numerical quadrature, not experimental error bars.
[WARNING] temperature: Axis temperatures are energy-matched bound-state mixture proxies; survivor cooling of undriven axes may
result from preferential loss.
```